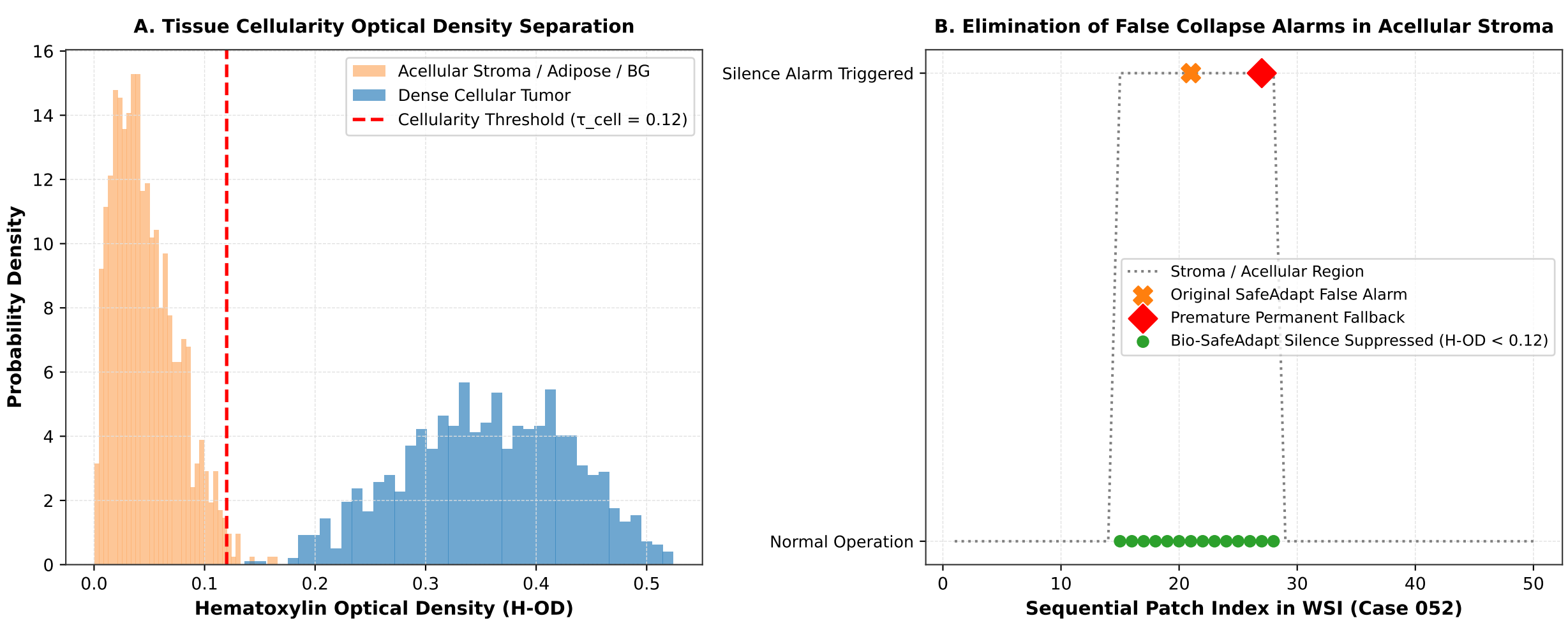
A. Tissue Cellularity Optical Density Separation
Probability Density
Hematoxylin Optical Density (H-OD)
Acellular Stroma / Adipose / BG
Dense Cellular Tumor
Cellularity Threshold (τ_cell = 0.12)
0
2
4
6
8
10
12
14
16
0.0
0.1
0.2
0.3
0.4
0.5
B. Elimination of False Collapse Alarms in Acellular Stroma
Silence Alarm Triggered
Normal Operation
Sequential Patch Index in WSI (Case 052)
Stroma / Acellular Region
Original SafeAdapt False Alarm
Premature Permanent Fallback
Bio-SafeAdapt Silence Suppressed (H-OD < 0.12)
0
10
20
30
40
50